

Energy Trading

Workshop reference pack

The energy_trading dataset at a glance

A market-wide trade tape: 7 dimensions, 3 facts, ~1.0M trades across 32 companies and 122 traders. This pack collects the diagrams and the KPI cheat-sheet for offline use during the workshop.

Contents

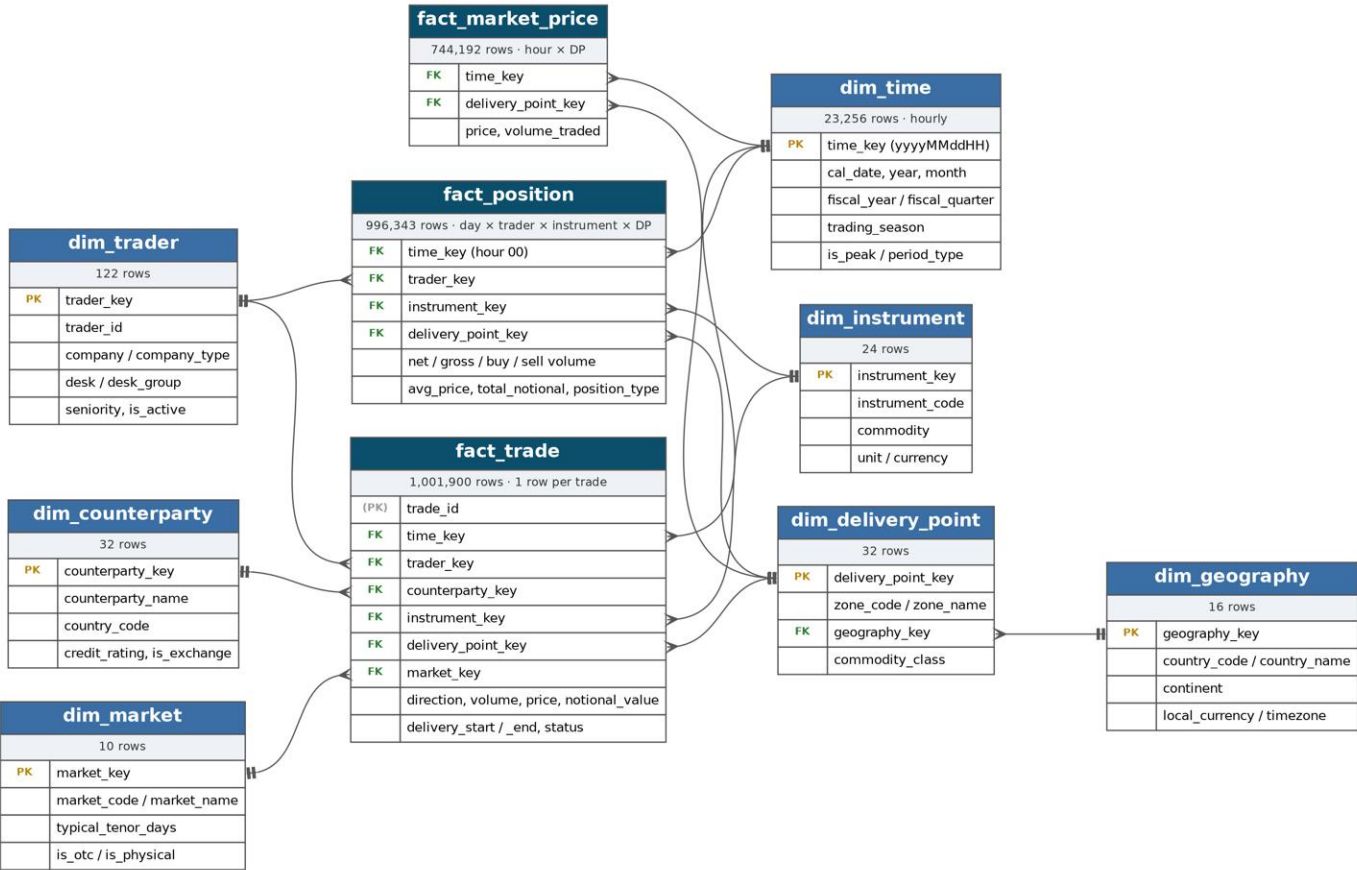
1. ER diagram — the full star schema
2. Common join paths
3. KPI map — grouped by business question
4. Whose data is this? (market-wide, not one company)
5. The surveillance story

Key framing for participants

This is exchange / regulator-style data — every participant, every deal.

Money totals mix EUR/GBP/USD; exclude cancelled trades before summing.

Er Energy Trading

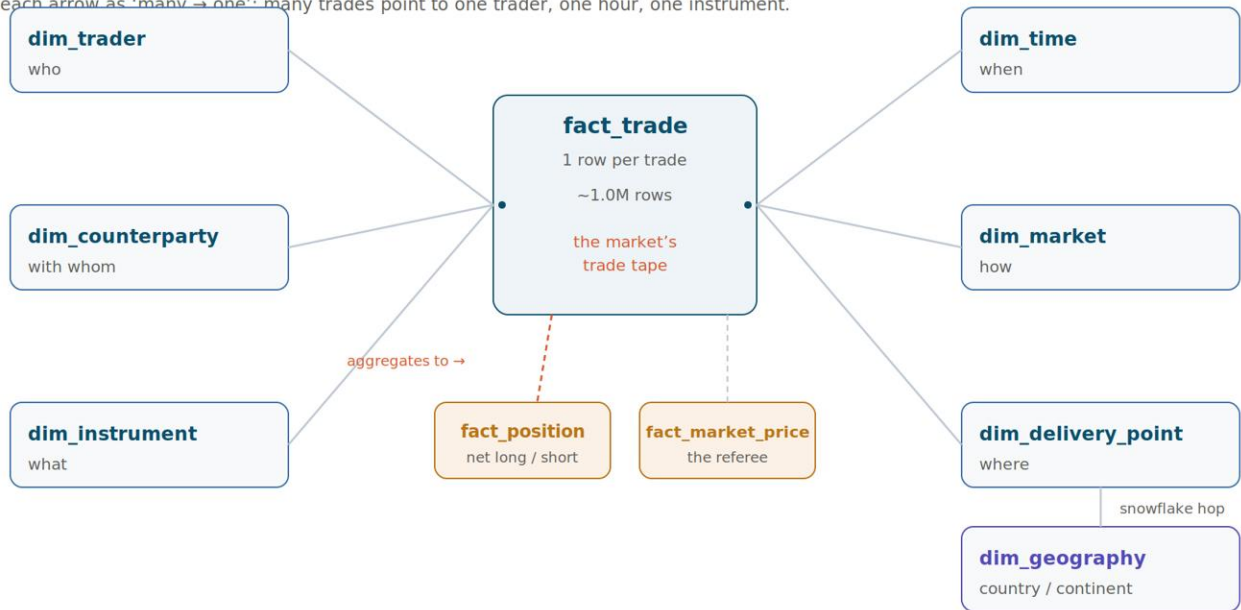


Join Paths

Common join paths

How the energy_trading tables connect for analysis

Read each arrow as 'many $\rightarrow$ one': many trades point to one trader, one hour, one instrument.



Tip: to reach country/continent, join fact_trade $\rightarrow$ dim_delivery_point $\rightarrow$ dim_geography (two hops).

Kpi Map

KPI map

What you can ask — grouped by the business question

1. Activity how busy is the market? Trade count, total volume, notional, unique traders/counterparties pre-built	2. Price what price are deals struck at? Average price, VWAP (volume-weighted), min / max price pre-built
3. Direction who is buying vs selling? Buy vs sell volume & notional, net long / short position pre-built	4. Money market-wide sell-side value Revenue, trading profit — mixed currencies, handle with care caveat
5. Trend how is it changing over time? Rolling 6M / 12M volume, month-over-month growth % pre-built	6. Surveillance is anyone behaving oddly? Price vs market %, oversized clips — the demo highlight derive

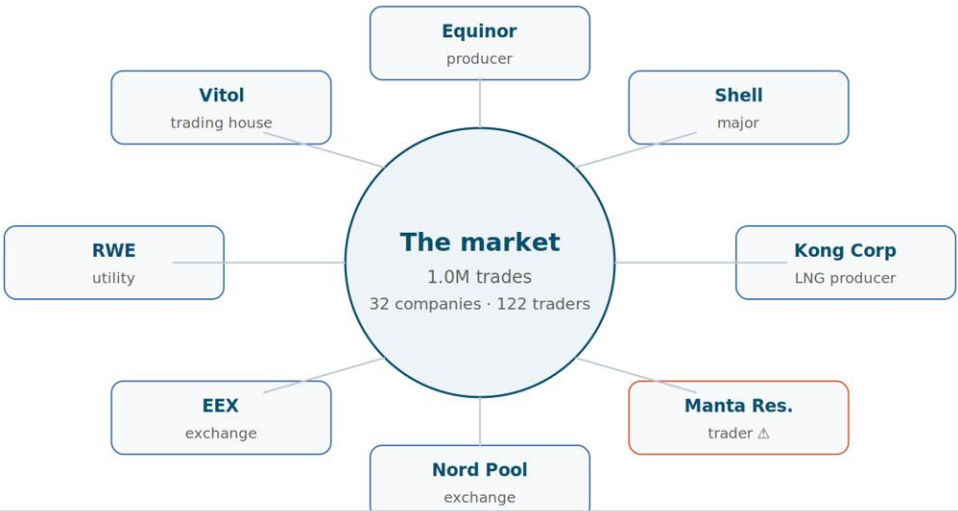
This is market-wide data (every participant). 'Revenue' = total across all 32 companies, not one firm's P&L.

Market Structure

Whose data is this?

A market-wide trade tape — not one company's books

Every row is one deal between two of the market's participants. There is no single 'us'.



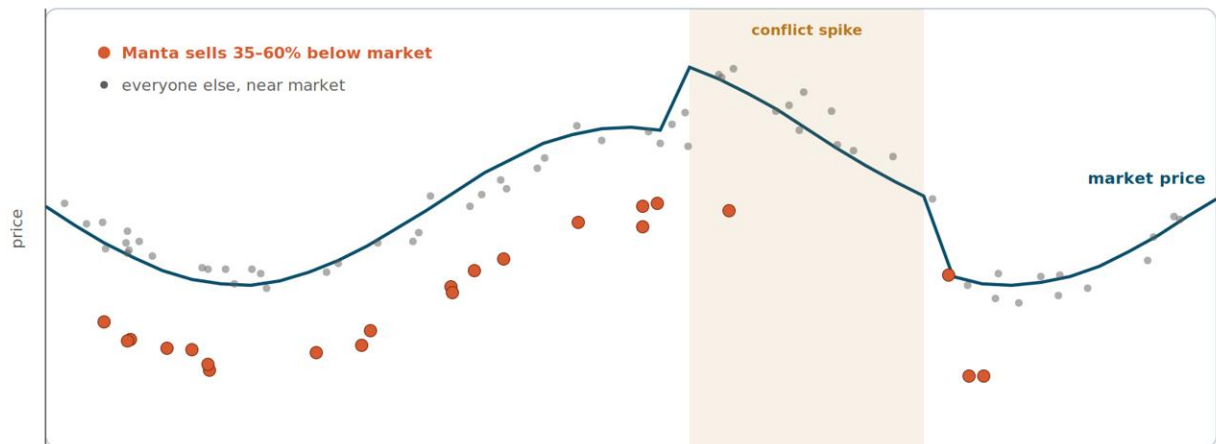
Same company can be the trader on one row and the counterparty on another — the classic sign of exchange / regulator data.

Surveillance Story

The surveillance story

Why B.Rock at Manta Resources gets caught

The reference price (fact_market_price) is the yardstick. Compare each trade's price against it.



Genie question: 'Which trader consistently sold furthest below the market price, and did it worsen during spike windows?'

Answer: B.Rock Van Guard (Manta Resources) — the built-in 'panic the market' seller. This is the demo's payoff.